

How much has power consumption increased because Google has included AI-answers in their search engine?

Posted by u/TopptrentHamster • 0 points • 3 comments

Most people do not turn this option off or include "-ai" in their search to disable the AI-answer. AI is known to be very energy intensive. What has including AI in every search done to power consumption world wide?

Comments

u/MuJartible • 1 points • 2025-06-16 20:36 UTC

>Most people do not turn this option off

How can I do that?

u/Lumpy-Notice8945 • 2 points • 2025-06-16 12:21 UTC

> AI is known to be very energy intensive.

Its really not, at least using it is not energy intensive, training it is what is consuming so much power.

So a single google search does not use much more electricity with the ai.

u/GooglyEyedGramma • 1 points • 2025-06-18 10:33 UTC

Doesn't running it also use a lot of energy though? Wasn't there a whole thing about how OpenAI and others were using massive amounts of energy per query? Is it because the models are being trained while in use?

u/Lumpy-Notice8945 • 1 points • 2025-06-18 11:05 UTC

I have heard this claim multiple times in reddit comments but i dont know why people belive that.

Deepseek (the recent chinese LLM) runs on a phone locally, a phone does not even have enough power stored in its battery. I can generate full HD images on my gaming PC at home with a decent GPU in about one or two minutes, my PC has a 800W power supply.

I have not found any actual numbers about chatGPT/OpenAI but i really doubt they somehow use a thousand times or more of what any other model uses(and even that

would not come close to some claims i have heard like "as much as a country")

And yes they use a oot of energy for training, but even these numbers are often exaggerated or just plain wrong, most "sources" i have seen are news articles that dont really provide a proper source, they either add together all the models the AI industry uses and ofc thats a lot, but then you need to compare that to the whole robot industry or all of the websites to be fair and then its not that much anymore.

I think many people just confuse crypto with AI, because ctypto(proof of work mining) uses up a lot of electricity.

u/GooglyEyedGrama • 2 points • 2025-06-18 11:51 UTC

It seems you might be right: <https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use>, though, I haven't yet read it completely, and just skimmed it since I'm at work.

u/Lumpy-Notice8945 • 1 points • 2025-06-18 12:18 UTC

I would not trust these numbers too much either, there is a lot of super vague estimations and just guessing in there too that could be off by a factor of 10 or more.

Like i cant belive an AI uses only two FLOPs per parameter that sounds so far of what i would expect that i just cant tell how they get to that number at all, a single instruction on one core of a GPU is basically already one FLOP because thats what GPUs are buid for, lots and lots of floating point operations in parralel. And runing a AI should be some kind of matrix multiplication for each layer and the matrix should have the size of the parameters.

But what tells a lot is that it seems like openAI runs their model on NVidia GPUs that use 700-1500W. Thats about the same a high end personal PC uses and it seems they dont need to combine clusters of multiple of these for a single output and it doesnt take more than a second to generate an output.

So these are not huge server farms that generate a single answer but more comparable to high end PCs. That sounds like if you play a game you use more electricity than if you ask chatGPT a cupple of questions.

u/GooglyEyedGrama • 2 points • 2025-06-18 12:35 UTC

Yeah, the numbers sound a bit too optimistic (contrary to what they actually say), but the final

number does look fairly realistic. I guess the optimistic assumptions get toned down by the pessimistic ones. Even if the number is higher, it's still fairly small tbh. Like you said, doubt that it's the worst part, and that the impact is way bigger than everything we already have, like google search and things like that.

u/Exactly65536 • 1 points • 2025-06-16 12:20 UTC

Running AI is relatively cheap (although might still matter at scale), it's training that consumes a lot of energy.

Estimates have it that 1 query costs from 1 to 10 W*h (3600-36000 J).

Google runs 100k queries per second, so it's approximately extra 100 kWh - 1 MWh per each second.

Or just 360 - 3600 MW.

I imagine these answers are on a cheaper side as compared to "write a code for a new twitter", so I'd say 1/3 of a Gigawatt. About a third of a nuclear power plant's output.